Metric	Definition	Interpretation
Valid flips	$B(M(x)) = 1,\ B(M_{\mathcal{S},c}(x)) = 0$	Defense-succeeded cases
Recovery rate	$\mathbb{E}_{x \in \mathcal{V}} \mathbf{1}[B(M^{\mathrm{rec}}_{\mathcal{S},c}(x)) = 1]$	Behavior restored
Non-refusal recovery	$\mathbf{1}$[recovered output has no refusal/safety-cue opening]	Weak refusal-removal readout
Base-like recovery	$\mathbf{1}$[recovered output is close to the unclamped base response]	Fidelity to base non-refusal behavior
Activation drift	$\lVert E_{\ell,\mathcal{S}}(h^{\mathrm{rec}}_{\ell}(x)) - c_{\mathcal{S}} \rVert$	Movement of defended features
Zero-reactivation recovery	$\mathbf{1}[B(M^{\mathrm{rec}}_{\mathcal{S},c}(x)) = 1 \wedge \max_{i \in \mathcal{S}} z^{\mathrm{rec}}_{\ell,i}(x) \le \eta]$	Recovery without reopening selected features
Defended-feature drift	$\lVert z^{\mathrm{rec}}_{\mathcal{S}} - z^{\mathrm{def}}_{\mathcal{S}} \rVert_2 / (\lVert z^{\mathrm{def}}_{\mathcal{S}} \rVert_2 + \epsilon)$	Movement from post-clamp feature state
Clamp-floor violation	$\lVert [z^{\mathrm{def}}_{\mathcal{S}} - z^{\mathrm{rec}}_{\mathcal{S}}]_+ \rVert_2 / (\lVert z^{\mathrm{def}}_{\mathcal{S}} \rVert_2 + \epsilon)$	Whether recovery lowers clamped refusal features
Relative perturbation norm	$\lVert \delta_x \rVert_F / (\lVert h^{\mathrm{def}}_{\ell}(x) \rVert_F + \epsilon)$	Scale of residual recovery path

Table 1: Metrics for post-intervention recovery. $\mathcal{S}$ denotes the defended SAE feature set and $c_{\mathcal{S}}$ denotes the defended feature values.

A Evaluation Protocol and Metrics

Valid-flip set. For a task-specific behavior predicate B, we define the valid-flip set as

$$\mathcal{V} = \{x : B(M(x)) = 1,\ B(M_{\mathcal{S},c}(x)) = 0\}.$$

Thus, an example is included only if the original model exhibits the target behavior and the defended model no longer does after the SAE intervention is applied. This conditioning ensures that recovery is evaluated only when there is a behavior that the intervention has actually suppressed.

Recovery metrics. We report recovery rate as the primary behavioral metric and defended-feature drift as the primary feature-preservation metric. For binary outcomes, recovery is counted only on the valid-flip set $\mathcal{V}$. For refusal recovery, we additionally report base-like recovery, defended-feature drift, and clamp-floor violation. Base-like recovery measures whether the recovered response remains close to the original non-refusal response, while clamp-floor violation measures whether recovery lowers the clamped refusal features below their post-clamp value.

Non-refusal and base-like recovery. For refusal experiments, we report two behavioral metrics. *Non-refusal recovery* is a weak readout that checks whether the recovered response avoids explicit refusal or safety-cue openings such as "I cannot" or "I can't help". This measures whether the clamp-induced refusal behavior has been removed, but it does not guarantee that the response is coherent: because the active clamp and recovery update can perturb the latent trajectory, a non-refusal output may still be degenerate, malformed, or unrelated to the original answer. We therefore also report *base-like recovery*. Since recovery is optimized toward the unclamped base model's non-refusal response, base-like recovery asks whether the recovered output remains close to that base response rather than merely avoiding refusal prefixes. Thus, non-refusal recovery measures refusal removal, while base-like recovery measures response fidelity.

Uncertainty estimates. For binary recovery outcomes, we report Wilson 95% confidence intervals. For continuous quantities such as defended-feature drift, clamp-floor violation, and relative δ_x norm, we report bootstrap confidence intervals over valid examples when saved per-example values are available.

B Additional Standard-Task Results

B.1 TPP target-mean results

Table 2 reports the dataset-level target means for the official layer-5 TPP benchmark. Values are averaged over target classes within each dataset. The unconstrained variant measures how much target information remains recoverable after official SAE zero-ablation, while the encoder-projected variant tests whether recovery can persist when updates are projected away from the defended SAE encoder directions. Reactivation, activation drift, and zero-reactivation recovery are measured post hoc rather than directly optimized in the main TPP runs.

Supplementary visualization. Figure 7 provides an additional visualization of the recovery–reactivation behavior across official layer-5 TPP targets. The main text uses Figure 2 for the primary trade-off; this appendix figure and Table 2 provide the full target-mean summary.

Dataset	Rec. None	Rec. Enc.	React. None	React. Enc.	Drift None	Drift Enc.	Zero-React None	Zero-React Enc.
Bias in Bios 1	0.831	0.754	0.011	0.002	0.248	0.059	0.128	0.715
Bias in Bios 2	0.803	0.745	0.011	0.002	0.041	0.034	0.140	0.703
Bias in Bios 3	0.720	0.629	0.012	0.002	0.050	0.040	0.107	0.710
Amazon Reviews	0.924	0.869	0.015	0.002	0.039	0.022	0.035	0.594
All targets (unweighted)	0.819	0.749	0.013	0.002	0.094	0.039	0.103	0.680

Table 2: Target-mean comparison between unconstrained and encoder-projected recovery on official layer-5 TPP. Encoder projection reduces mean defended-feature reactivation and activation drift while preserving substantial valid-flip recovery.

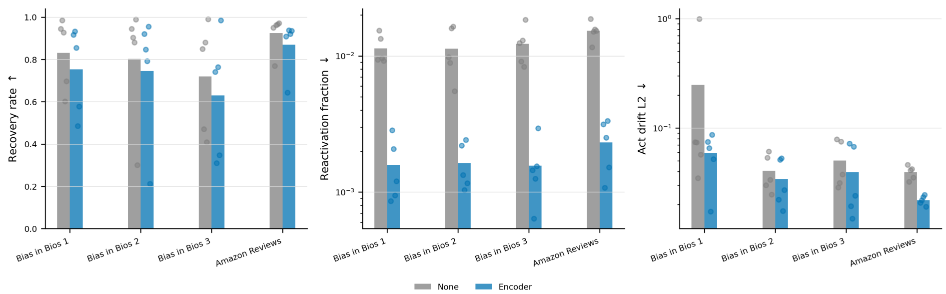

Figure 7: Supplementary TPP recovery summary across official layer-5 targets.

C Additional Refusal Validity Checks

Strict valid-filtering protocol. Our refusal recovery experiments are conditioned on prompts for which the active refusal feature intervention actually induces refusal. This is necessary because recovery is only meaningful when there is a suppressed behavior to recover from. For the base-response recovery setting, a prompt is valid only if the unclamped model gives a non-refusal response without safety-cue openings, and the clamped model gives a refusal under the same detector.

Let $R(y)$ denote the automatic refusal detector and $S(y)$ denote a safety-cue detector. A prompt x_i is counted as valid when

$$\mathrm{valid}_i = \mathbf{1}\left[\neg R(y_i^{base}) \wedge \neg S(y_i^{base}) \wedge R(y_i^{clamp})\right].$$

This criterion ensures that the recovery target is a non-refusal base response and that the SAE clamp has actually moved the model into a refusal-like behavior.

Feature-set-specific valid filtering. We use feature-set-specific valid filtering rather than assuming that a nominal refusal feature set is valid by construction. On the same 520 AdvBench prompts, different refusal feature sets can induce substantially different clamp behavior. This supports our design choice to evaluate recovery only after verifying that the selected clamp actually suppresses the target behavior.

Interpretation. The base generation is computed without any feature clamp. Thus, the difference in valid sample count is not caused by the feature set changing the base model behavior; it is caused by the clamp having substantially different behavioral effect. The `benchmark_our` feature set induces refusals on 203 prompts and yields 24 strict valid recovery cases. In contrast, `benchmark_la` induces refusals on only 48 prompts and has only two strict valid cases.

Manual inspection further shows that the two strict `benchmark_la` cases are degenerate base generations dominated by repeated punctuation or tokens rather than usable non-refusal answers. This failure mode is not captured by a substring refusal detector, which only checks for refusal and safety-cue phrases. Consequently, `benchmark_la` does not provide a reliable valid set for base-response recovery under this protocol. We therefore run the main recovery experiments only on the `benchmark_our` feature set.

Dataset	Strict valid	Non-ref. recovery	Base-like recovery	Strict base-like	Drift	Floor viol.
AdvBench	24	23/24	15/24	12/24	0.131	0.127
HarmBench-Test	43	43/43	19/43	14/43	0.108	0.102

Table 3: Cross-dataset strict-valid refusal recovery. HarmBench-Test uses the same active SAE clamp, strict-valid filtering, Jacobian projection, and post-hoc defended-feature evaluator as the AdvBench main experiment.

Output source	Non-refusal candidate	Partial refusal	Full refusal / degenerate
Base	23	0	1
Clamp	0	0	24
Jacobian recovery	19	4	1

Table 4: Automatic opening-category sanity check on the 24 AdvBench strict-valid examples. The active clamp consistently moves outputs into full-refusal openings, while Jacobian recovery mostly returns to non-refusal-candidate openings. Manual labels are pending, so this table should be interpreted as an automatic diagnostic rather than an independent human evaluation.

D Cross-Dataset Refusal Recovery

HarmBench-Test protocol. To check whether the refusal recovery result is specific to the AdvBench strict-valid subset, we repeat the same strict-valid filtering and Jacobian-projected recovery protocol on HarmBench-Test. We use the same model, SAE feature set, clamp value, refusal detector, projection method, and defended-feature evaluator as in the AdvBench refusal experiment. As in the main experiment, a prompt is included only if the base model gives a non-refusal response without a safety-cue opening and the active SAE clamp changes the output into a refusal-like response.

Results. Table 3 shows that the recovery–preservation pattern is not limited to AdvBench. On HarmBench-Test, the strict filter yields 43 valid examples out of 159 prompts. Jacobian-projected recovery restores non-refusal behavior on all 43 valid examples, with 19/43 base-like recoveries and 14/43 strict-base-like recoveries. The mean defended-feature drift is 0.108 and the mean clamp-floor violation is 0.102, close to the AdvBench defended-feature preservation level.

E Opening-Category Sanity Check

Because non-refusal recovery only checks for the absence of explicit refusal openings, we include an additional opening-category diagnostic. A recovered output may avoid prefixes such as "I cannot" while still being a partial refusal, a degenerate generation, or a low-fidelity response. We therefore inspect coarse opening categories to check whether the recovery signal is dominated by obvious detector artifacts. This diagnostic is not a substitute for a full independent human evaluation; manual labels are still pending. For safety reasons, full harmful completions are not saved or reported.

HarmBench-Test opening categories. On the 43 HarmBench-Test strict-valid examples, the recovered openings are categorized as 37 non-refusal-or-other openings, 4 degenerate/repeated openings, and 2 partial-refusal or safety-cue openings. This suggests that the 43/43 automatic non-refusal recovery result is not solely driven by a single opening pattern, although a full manual audit remains necessary for stronger claims about response quality.

F Redacted Qualitative Example

To illustrate the distinction between refusal removal and base-like recovery, Table 5 provides a redacted side-by-side example from the refusal recovery case study, including the base response, the clamp-induced refusal, and the recovered response. The full prompt and harmful procedural details are intentionally omitted for safety. This example is not used as an additional quantitative result; it is included only to clarify how the base, clamped, and recovered trajectories differ under our metrics.